

30. 25.3 g of sodium hydroxide was mixed with 9.60 g of magnesium chloride and then placed into enough water to produce 8.45 L of solution. Calculate:
- the $[\text{OH}^-]$ ion concentration of the resulting solution;
 - the $[\text{H}^+]$ ion concentration of the resulting solution;
 - the pH of the resulting solution.
31. The standardisation of an approximately 0.05 mol L^{-1} HCl solution with 20.0 mL aliquots of a $0.0997 \text{ mol L}^{-1}$ Na_2CO_3 solution produced the following table of results:

	Titration Number				
	Rough	1	2	3	4
Final Reading (mL)	45.6	43.00	46.40	44.80	49.30
Initial Reading (mL)	2.0	1.00	5.00	3.50	7.90
Titration Volume (mL)					

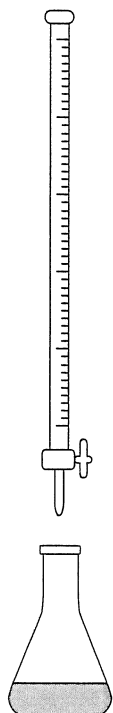
- Write a balanced equation for the reaction.
 - Calculate the number of moles of Na_2CO_3 used in each titration.
 - Determine the average titre volume of HCl used.
 - Determine the concentration of the HCl solution.
32. The sodium hydroxide concentration of an oven cleaning solution was determined by the steps outlined below.
- 25.00 g of the oven cleaner was dissolved in 100 mL of distilled water.
 - This solution was then added to a 250.0 mL volumetric flask and diluted by the addition of enough distilled water to correctly fill the flask.
 - A 24.00 mL aliquot was removed from the volumetric flask and diluted by addition to enough water to fill another 250.0 mL volumetric flask.
 - 20.00 mL samples of this diluted solution were titrated against a standardised solution of HCl. An average of 29.80 mL of the 0.107 mol L^{-1} HCl was used.
- Assuming that the only basic substance in the oven cleaner was sodium hydroxide, determine the sodium hydroxide concentration of the diluted solution produced in step III.
 - Calculate the percentage, by mass, of NaOH in the original oven cleaning solution.

33. Car battery acid contains approximately $4.500 \text{ mol L}^{-1} \text{ H}_2\text{SO}_4$. A student carried out a series of titrations to determine the accurate concentration of H_2SO_4 in battery acid. The student followed the following steps.

- STEP I Take 20.0 mL of the battery acid and dilute to 1.00 L in a volumetric flask.
- STEP II Fill a burette with the diluted battery acid.
- STEP III Place 25.0 mL of standardised Na_2CO_3 ($0.0939 \text{ mol L}^{-1}$) into a conical flask and add several drops of methyl orange.
- STEP IV Conduct enough titrations to get results which are consistent to $\pm 0.1 \text{ mL}$.

Calculate the concentration of the undiluted battery acid if the average titration volume obtained in step IV was 24.7 mL.

34. When the equivalence point in a titration between a standardised $0.100 \text{ L}^{-1} \text{ HCl}$ solution and an approximately $0.10 \text{ mol L}^{-1} \text{ NaOH}$ has been reached, 24.90 mL of the acid had been added from a burette to 20.0 mL of NaOH in a conical flask. What would be the change in pH caused by the addition of another 0.100 mL drop of acid into the solution in the conical flask?



27.

$$(a) \quad m(\text{battery acid}) = D \times V \\ = 1.224 \times 2.00 \times 10^3 = 2440 \text{ g} \\ m(\text{H}_2\text{SO}_4) = 35.0\% \text{ of } m(\text{battery acid})$$

$$= 2440 \times \frac{35.0}{100} = 854 \text{ g}$$

$$n(\text{H}_2\text{SO}_4) = \frac{m}{M} = \frac{854}{98.086} = 8.71 \text{ mol}$$

$$n(\text{H}_2\text{SO}_4) \text{ per cell} = 8.71 \div 6 = 1.45 \text{ mol}$$

28.

$$(a) \quad \min [\text{H}^+] = \text{inv log } (-7.8)$$

$$= 1.58 \times 10^{-8} \text{ mol L}^{-1}$$

$$(b) \quad \text{no. } \text{H}_3\text{O}^+ \text{ ions} = c \times V \times 6.022 \times 10^{23} \\ = 1.58 \times 10^{-8} \times 3.50 \times 10^4 \times 6.022 \times 10^{23} \\ = 3.34 \times 10^{20} \text{ ions}$$

$$(c) \quad \text{If } \text{pH} = 7.2, [\text{H}^+] = \text{inv log } (-7.2)$$

$$= 6.31 \times 10^{-8} \text{ mol L}^{-1}$$

$$n(\text{acid}) \text{ in pool if } \text{pH} = 7.8 = cV$$

$$= 1.58 \times 10^{-8} \times 3.50 \times 10^4 = 5.53 \times 10^{-4} \text{ mol}$$

$$n(\text{acid}) \text{ in pool if } \text{pH} = 7.2 = cV$$

$$= 6.31 \times 10^{-8} \times 3.50 \times 10^4 = 2.21 \times 10^{-3} \text{ mol}$$

$$n(\text{HCl}) \text{ to be added}$$

$$= 2.21 \times 10^{-3} - 5.53 \times 10^{-4}$$

$$= 1.657 \times 10^{-3} \text{ mol}$$

$$(d) \quad V(\text{conc. HCl}) = \frac{n}{c} = \frac{1.657 \times 10^{-3}}{12}$$

$$= 1.38 \times 10^{-4} \text{ L } (0.138 \text{ mL})$$

$$29. \quad n(\text{Mg}(\text{OH})_2) = \frac{m}{M} = \frac{30.0}{58.316} = 0.514; \text{ mol}$$

$$n(\text{OH}^-) = 2 \times 0.514 = 1.029$$

$$n(\text{Al}(\text{OH})_3) = \frac{m}{M} = \frac{30.0}{78.004} = 0.385; \text{ mol}$$

$$n(\text{OH}^-) = 3 \times 0.385 = 1.154 \text{ mol}$$

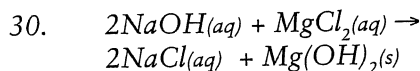
$$n(\text{OH}^-) \text{ in } 375 \text{ mL} = 1.029 + 1.154 = 2.18 \text{ mol}$$

$$n(\text{OH}^-) \text{ in } 10.0 \text{ mL} = 2.18 \times \frac{10}{375} = 0.0582 \text{ mol}$$

$$\therefore n(\text{HCl}) \text{ neutralised} = n(\text{OH}^-) \text{ in } 10.0 \text{ mL} \\ = 0.0582$$

$$V(\text{HCl}) \text{ neutralised}$$

$$= \frac{n}{c} = \frac{0.0582}{\text{inv log } (-0.9)} = 0.462 \text{ L}$$



$$n(\text{NaOH}) = \frac{m}{M} = \frac{25.3}{39.998} = 0.633 \text{ mol}$$

$$n(\text{MgCl}_2) = \frac{m}{M} = \frac{9.60}{95.2} = 0.101 \text{ mol}$$

(Limiting Reagent)

$$n(\text{OH}^-) \text{ consumed} = 2n(\text{MgCl}_2) = 0.202 \text{ mol}$$

$$(a) \quad n(\text{OH}^-) \text{ remaining in solution} \\ = 0.633 - 0.202 = 0.431 \text{ mol} \\ c(\text{OH}^-) \text{ remaining in solution}$$

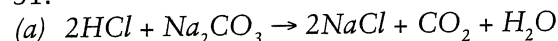
$$= \frac{n}{V} = \frac{0.431}{8.45} = 0.0510 \text{ mol L}^{-1}$$

$$(b) \quad c(\text{H}^+) = \frac{1.00 \times 10^{-14}}{c(\text{OH}^-)} = \frac{1.00 \times 10^{-14}}{0.0510}$$

$$= 1.96 \times 10^{-13} \text{ mol L}^{-1}$$

$$(c) \quad \text{pH} = -\log [\text{H}^+] = 12.7$$

31.



$$(b) \quad n(\text{Na}_2\text{CO}_3) = cV = 0.997 \times 0.0200$$

$$= 1.994 \times 10^{-3} \text{ mol}$$

$$(c) \quad \text{average } V(\text{HCl}) =$$

$$(\text{titration 2} + \text{titration 3} + \text{titration 4}) \div 3$$

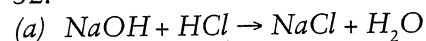
$$= 41.4 \text{ mL}$$

$$(d) \quad n(\text{HCl}) = 2n(\text{Na}_2\text{CO}_3) = 3.988 \times 10^{-3} \text{ mol}$$

$$c(\text{HCl}) = \frac{n}{V} = \frac{3.99 \times 10^{-3}}{0.0414}$$

$$= 0.0963 \text{ mol L}^{-1}$$

32.



$$n(\text{HCl}) = cV = 0.107 \times 0.0298$$

$$n(\text{HCl}) = 3.189 \times 10^{-3} \text{ mol}$$

$$n(\text{NaOH}) = n(\text{HCl}) = 3.189 \times 10^{-3} \text{ mol}$$

$$c(\text{dilute NaOH}) = \frac{n}{V} = \frac{3.189 \times 10^{-3}}{0.0200}$$

$$= 0.159 \text{ mol L}^{-1}$$

(b) (Oven cleaner was diluted twice: in steps II and III)

 $n(\text{NaOH})$ in 250 mL of dilute solution from step III

$$= cV = 0.159 \times 0.250$$

$$n(\text{NaOH}) \text{ in } 250 \text{ mL from step III}$$

$$= 0.0399 \text{ mol}$$

$$\therefore n(\text{NaOH}) \text{ in } 24.0 \text{ mL from step II}$$

$$= 0.0399 \text{ mol}$$

$$c(\text{NaOH}) \text{ from step II} = \frac{0.0399}{0.0240}$$

$$= 1.66 \text{ mol L}^{-1}$$

$$\therefore n(\text{NaOH}) \text{ in all of solution in step II}$$

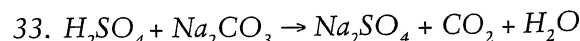
$$= cV = 1.66 \times 0.250 = 0.415 \text{ mol}$$

$$m(\text{NaOH}) \text{ in all of solution in step II}$$

$$= nM = 0.415 \times 39.998 = 16.6 \text{ g}$$

$$\% \text{ NaOH in drain cleaner}$$

$$= \frac{16.6}{25.0} \times \frac{100}{1} = 66.4\% \text{ (w/w)}$$



$$n(\text{Na}_2\text{CO}_3) = cV = 0.0939 \times 0.025 = 2.35 \times 10^{-3}$$

$$\therefore n(\text{dilute H}_2\text{SO}_4) = n(\text{Na}_2\text{CO}_3) = 2.35 \times 10^{-3}$$

$$c(\text{dilute H}_2\text{SO}_4) = \frac{n}{V} = \frac{2.35 \times 10^{-3}}{0.0247} = 0.0950 \text{ mol L}^{-1}$$

$$n(\text{dilute H}_2\text{SO}_4) \text{ in } 1.00 \text{ L} = cV = 0.0950 \times 1.00 = 0.0950 \text{ mol}$$

$$\therefore n(\text{conc. H}_2\text{SO}_4) \text{ in } 20.0 \text{ mL} = n(\text{dilute H}_2\text{SO}_4) \text{ in } 1.00 \text{ L} = 0.0950 \text{ mol}$$

$$c(\text{battery acid}) = \frac{n}{V} = \frac{0.0950}{0.0200} = 4.75 \text{ mol L}^{-1}$$

34. At equivalence point the solution is neutral i.e. $[\text{H}^+] = 1.00 \times 10^{-7} \text{ mol L}^{-1}$
 $V(\text{solution}) = 0.0249 + 0.0200 = 0.0449 \text{ L}$
 $\therefore n(\text{H}^+) \text{ at equivalence point} = cV = 1.00 \times 10^{-7} \times 0.0449 = 4.49 \times 10^{-9} \text{ mol}$
 $n(\text{H}^+) \text{ in } 1 \text{ drop} = cV = 0.100 \times 0.0001 = 1.00 \times 10^{-5} \text{ mol}$
 $\therefore n(\text{H}^+) \text{ in solution after } 1 \text{ drop added} = 4.49 \times 10^{-9} + 1.00 \times 10^{-5} = 1.00 \times 10^{-5} \text{ mol}$
- $$c(\text{H}^+) = \frac{n}{V} = \frac{1.00 \times 10^{-5}}{0.0449} = 2.23 \times 10^{-4} \text{ mol L}^{-1}$$
- new pH = $-\log(2.23 \times 10^{-4}) = 3.65$
 $\therefore \text{change in pH} = 7.00 - 3.65 = 3.35$

For the Experts

- 35.
- (a) $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
- (b) $n(\text{HCl}) = cV = 0.136 \times 0.0226 = 3.07 \times 10^{-3} \text{ mol}$
- (c) $n(\text{Na}_2\text{CO}_3) \text{ in } 20 \text{ mL} = 1/2 n(\text{HCl}) = 1.54 \times 10^{-3} \text{ mol}$
- (d) $n(\text{Na}_2\text{CO}_3) \text{ in } 500 \text{ mL of solution} = \frac{500}{20} \times 1.54 \times 10^{-3} = 3.84 \times 10^{-2} \text{ mol}$
- $m(\text{Na}_2\text{CO}_3) \text{ in } 500 \text{ mL of solution} = nM = 3.84 \times 10^{-2} \times 105.99 = 4.07 \text{ g}$
 $\therefore m(\text{H}_2\text{O}) \text{ in } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = m(\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}) - m(\text{Na}_2\text{CO}_3) = 10.96 - 4.07 = 6.89 \text{ g}$
- (e) $n(\text{H}_2\text{O}) \text{ in } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = \frac{6.89}{18.016} = 0.382 \rightarrow \frac{n(\text{H}_2\text{O})}{n(\text{Na}_2\text{CO}_3)} = \frac{0.382}{3.84 \times 10^{-2}} \text{ mol}$
i.e. $n(\text{H}_2\text{O}) = 10 \times n(\text{Na}_2\text{CO}_3)$
 $\therefore \text{correct formula} = \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

CHP 3: OXIDATION AND REDUCTION

Chapter Questions

3.1

- (a) $\text{Mg(s)} + \text{Cl}_2(\text{g}) \rightarrow \text{MgCl}_2(\text{s})$
(b) Mg (c) Mg (d) Cl_2

3.2 True: oxidant removes electrons from reductant, e.g. $2\text{Fe} + \text{O}_2 \rightarrow 2\text{FeO}$
 O_2 removes electrons from Fe to form O^{2-} and Fe^{2+} .

3.3

- | | |
|----------|----------|
| (a) (+6) | (b) (+6) |
| (-2) | (-2) |
| (c) (+5) | (d) (-3) |
| (-2) | (+1) |
| | (-1) |
| (e) (+1) | (f) (+2) |
| (+4) | (+4) |
| (-2) | (-2) |

3.4

- (a) Redox (b) Not Redox
(c) Not Redox (d) Redox

3.5

- (a) oxidation: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2e^-$
reduction: $2\text{H}^+(\text{aq}) + 2e^- \rightarrow \text{H}_2(\text{g})$
(b) oxidation: $\text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{s}) + 2e^-$
reduction: $\text{S(s)} + 2e^- \rightarrow \text{S}^{2-}(\text{s})$

3.6

- (a) $\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5e^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O(l)}$
(b) $\text{H}_2\text{C}_2\text{O}_4(\text{aq}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2e^-$
(c) $\text{CH}_3\text{CH}_2\text{OH(l)} \rightarrow \text{CH}_3\text{CHO(l)} + 2\text{H}^+(\text{aq}) + 2e^-$

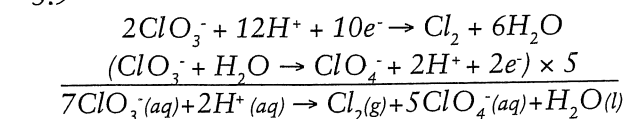
3.7

- (a) $(\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5e^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O(l)}) \times 2$
 $(\text{H}_2\text{S(g)} \rightarrow \text{S(s)} + 2\text{H}^+(\text{aq}) + 2e^-) \times 5$
 $2\text{MnO}_4^-(\text{aq}) + 5\text{H}_2\text{S(g)} + 6\text{H}^+(\text{aq}) \rightarrow 2\text{Mn}^{2+}(\text{aq}) + 5\text{S(s)} + 8\text{H}_2\text{O(l)}$
(b) $\text{ClO}^- + 2\text{H}^+ + 2e^- \rightarrow \text{Cl}^- + \text{H}_2\text{O}$
 $\text{SO}_2 + 2\text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 4\text{H}^+ + 2e^-$
 $\text{ClO}^-(\text{aq}) + \text{SO}_2(\text{g}) + \text{H}_2\text{O(l)} \rightarrow \text{Cl}^-(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq})$

3.8

- (a) $5\text{Fe}^{2+}(\text{aq}) + \text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) \rightarrow 5\text{Fe}^{3+}(\text{aq}) + \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O(l)}$
(b) $\text{Sn}^{2+}(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{OCl}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) + \text{Sn}^{4+}(\text{aq}) + \text{H}_2\text{O(l)}$

3.9



3.10

